

GIZMO Ground-Reference Monitoring

Concise Operations Guide

DUNE Far Detector · Connection, commissioning, and monitoring

Condensed from the public review guide dated August 18, 2026. Integration and commissioning draft; technical and electrical-safety approval required before field use.

Never lift the safety ground. Follow the released grounding drawings, ORC, LOTO, hazard analysis, and controlled procedure. Only authorized electrical personnel may change fixed grounds, CTs, saturable-inductor assemblies, protection, or welding returns. SOP004 authorizes GIZMO installation and use. Follow its current approved revision and the applicable work plan.

1. Purpose and operating basis

Detector Ground comprises the cryostat and intentionally bonded detector structures. Cavern/Building Ground comprises bonded infrastructure. The approved saturable-inductor path connects them for electrical safety while limiting small AC noise currents.

GIZMO injects a known AC stimulus between the references. A Pearson CT measures its return through the intended safety path. A parallel connection diverts current and changes the reported equivalent impedance or resistance. A low reading cannot locate the connection. An intentional bond also masks additional accidental shorts.

The Kria-generated OPC UA model 1.4.0, namespace `urn:fnal:gizmo`, is the machine-interface authority. Other hardware must conform; the SC/DPS ICD explains the contract. Endpoints, trust, roles, configuration, and field commands remain site-controlled. Legacy `RUN`, `load_f`, `set_th`, and `./GIZMO.elf` are not Kria commands.

The legacy excitation is a 1.4 kHz sine wave, with approximately 1 MS/s internal ADC/DAC operation. The maintained Kria default is 1436 Hz. The proposed 3 kHz Far Detector study value is not a commissioned setting. Select and approve the installed frequency by in-situ sweep and calibration.

Released drawings determine topology and component placement. The 2018 requirements specify at least two saturable inductors near the isolation transformer, separated by more than six feet; the 2025 laboratory ORC shows one MS6984 and one Pearson 2100. Resolve the installed arrangement through the released drawing.

2. Responsibilities and stop conditions

- **Electrical lead:** fixed grounds, CT, inductors/protection, rack bonds, approved fixture, and welding-return verification.
- **GIZMO expert:** chassis, image, analog chain, excitation, calibration, and alarm outputs.
- **SC/DPS expert:** network, time, history, replication, archive/restore checks, display, external state records, and coverage alarms.
- **Activity coordinator:** work-state declaration, planned-bond authorization and window, event records, and closeout.
- **Operator:** released checklist, live checks, event recording, and stop/escalation.

Stop before energizing or injecting if conductors are unidentified, the CT is bypassed, protection differs from the drawing, the rack safety bond is missing, image/configuration is unknown, or no responsible electrical/grounding contact is available.

Ground-affecting activity includes changes to bonds, welding returns, cable shields, pipes, scaffolding, test jumpers, grounding, or monitoring conductors. Pause/restart decisions follow activity and coverage, without a quota of stops per shift.

3. Prepare and connect

Repeat these checks after changes to the rack, topology, CT, inductor, firmware, or cryostat connection.

1. Confirm procedure scope, hazard analysis, released single-line/cable schedule, CT orientation, protection drawing, welding plan, and ORC for the installed location. Record equipment, software/FPGA, configuration hash, and calibration IDs; identify shift contacts.
2. Label both ground references, the CT conductor, both sides of the inductor assembly, stimulus injection point, and return/chassis bond. Verify drawings and text labels; color and photographs are insufficient.
3. With circuits safe and LOTO applied as required, inspect enclosures, connectors, strain relief, clearances, rack/chassis bonding, safety-path continuity, CT routing/security, unintended bridges, beacon visibility, and audible coverage.
4. With the rack de-energized, have the electrical worker verify the safety assembly. Connect CT output to labeled CT input; stimulus center to the approved detector-side point; return to the approved rack/infrastructure point. Use the approved cable and routing.
5. Connect beacon and audible outputs, assigned Kria Ethernet, and separate PDU management if required. Alarm-power connectors are not auxiliary supplies.
6. Connect the approved NRTL-listed converter and PDU branch circuit/adaptor; verify chassis rating and protective-earth continuity. The development-rack supply is 12 VDC, 5 A.
7. A second qualified person should check every connection against the drawing, read cable labels aloud, and record installer/verifier initials.

CT bypass or make-before-break work requires a separate released electrical procedure. Moving stimulus routing or injection points requires characterization review.

Post a revision-controlled rack-door reference: rack/device and ORC IDs, ground references, normal indications, alarm/quality/data-age interpretation, HIGH Z meaning, contacts, Daily Log location, procedure link, and “Never lift the safety ground.”

4. Power up and accept

1. Record external state as COMMISSIONING or NORMAL; confirm no undeclared welding, signed connections, storage headroom, logging/archive plan, provisioned replicas, and approved software/FPGA/configuration IDs. Keep excitation disabled until authorized.
2. Energize the PDU, then Kria supply. Check indicators and protective earth. De-energize immediately for abnormal sound, odor, heat, or smoke.
3. After the approved boot interval, verify `gizmo.target` and required services; record runtime/model versions. From the assigned SC/DPS host, verify endpoint, device identity, trust, and namespace resolution by URI.
4. Verify advancing UTC timestamps against the approved age limit. Start acquisition through the approved workflow; confirm waveform, frequency, amplitude limit, and calibration before the electrical lead authorizes stimulus.
5. Verify one-second fast snapshots, ten-second platform snapshots, advancing sequence numbers, monotonic timestamps, and no parse/quality flags.
6. Compare the stable measurement/range with the commissioned baseline. Verify display, edge-store progress, and all replica cursors.

Routine monitoring requires: Good measurement and `Alarm.Active` status; commissioned normal range/baseline; correct threshold/calibration; acceptable data age; active local history; explicitly recorded external work state. Historic laboratory values and the legacy ten-second time criterion do not establish Kria acceptance.

5. Commission the installed loop

Use a current procedure explicitly authorizing the installed topology and resistive/capacitive fixture. Legacy characterization required detector-isolation-transformer loads off, primary LOTO, and controlled detector-ground connections before inserting a decade box.

- Measure open/reference noise, known resistances through alarm conditions, representative capacitances, frequency response, and construction-equipment on/off effects.
- Select a frequency avoiding resonances/coherent noise, with monotonic short discrimination, stable I/Q, acceptable variance, and approved stimulus-current/touch/noise limits.
- Document threshold margins from normal noise and the maximum short resistance requiring detection. Do not adopt the legacy 100 Ω threshold or 5–6 Ω short reading without validation.
- Record sweep, fit, waveform, frequency, amplitude, uncertainty, calibration file/hash, and approver. Frequency, amplitude, CT, routing, inductor, or major bonding changes require review and normally re-characterization.
- Test beacon, audible output, SC/DPS alarm, and logging end to end. A failed alarm test blocks release.

6. Shift monitoring and records

At shift start, verify external activity state/source/time, current UTC data, Good alarm/measurement status, normal range/baseline, approved excitation/configuration/calibration/threshold, and historian/free-space/replica/archive health. Review the preceding 12 hours for steps, drift, gaps, or unclosed bond windows. Record start/end checks in Procure Daily Log > GIZMO.

The display should show range/resistance or HIGH Z, threshold, ≥ 12 -hour trend, I/Q/magnitude, source time/age/StatusCode, alarm/reason/latch/acknowledgements, configuration IDs, external activity/permit window, and system/logging health.

Activity state, permit/window, alarm inhibition, stimulus-enabled state, amplitude, and waveform are not supplied by the current OPC UA contract. Label their defined external sources.

Kria record	Cadence	Local retention
Fast measurement, alarm, lock-in, thermal, uptime, and sequence snapshot	1 second	14 days
Platform, OS, storage, process, filesystem, and network snapshot	10 seconds	30 days
Rollup: first, last, min, max, mean, count, and worst status	1 minute	365 days
Alarm, quality, clock, boot, network, service, and history transition	On change	1825 days

SQLite schema 2 retains scalar value, StatusCode, source time, receive time, and fast-data sequence/epoch; raw ADC/SDR frames are excluded. Planning estimates: 1.48 GB/month continuous capture; bounded policy approximately 1.26 GB in replay.

Create immutable archives using SQLite online backup or checkpointed export; do not copy an actively written WAL database. Record checksum, schema/runtime versions, model digest, and capture-path digest. Cursor replication transfers inserts and updated rollups, never deletions. Off-board replicas and PostgreSQL/Ignition history have independent retention, backup, and restore acceptance; permanent replicas may disable age pruning while retaining free-space guards.

Daily Log entries contain 6-, 12-, or 24-hour plots/statistics; shift checks; work-package/bond events; alarms, acknowledgements, investigation/disposition; and archive/checksum/configuration/calibration links. Procure is not the raw-data store.

7. Welding and planned bonds

1. **Before:** coordinator authorizes work package, location, lead, bond arrangement, and window; electrical lead verifies returns/bonds. Require Good measurement, Good clear alarm, and normal baseline for the approved pre-work interval. Log WELDING_PREP with UTC time and expected duration; verify continuous one-second capture.
2. **During:** log WELDING_ACTIVE at actual bond time. Verify the expected transition; stop for expert review if absent. Keep quality, age, health, and history visible; log subsequent changes. Impedance cannot establish isolation while bonded.

3. **If inhibited:** use only approved SC/DPS annunciation management; retain raw alarm/status/time and an external ALARM_INHIBITED record. Do not disconnect CT, stimulus, safety path, or monitor to silence an alarm.
4. **After:** work lead confirms stop and temporary leads/bonds removed or in specified final state. Log WELDING_BOND_REMOVED at removal. Require normal alarm/quality/range/baseline within the approved recovery window and stable post-work interval.
5. **Close:** if recovery fails, control the area and inspect recent work under the approved fault plan. Log WELDING_CLOSED only after electrical-lead/coordinator acceptance; attach before/after plots.

These event names are human-log labels, not OPC UA nodes. Approved procedures define timing and vocabulary. Unknown state, expired window, or mismatched bonds require treatment as a possible fault.

8. Alarm and coverage response

Observation	Required response
Good active alarm in NORMAL	Stop new ground-affecting activity; record UTC time/reason; notify coordinator and grounding/GIZMO expert; preserve data and investigate recent changes.
Good active alarm during approved bond window	Verify work package, location, bonds, and window; continue quality/age/history checks. Treat discrepancies as unexpected faults.
Good OutOfRange / HIGH Z, alarm clear	Record HIGH Z and compare with commissioned normal state. It is a valid range state; do not substitute a numeric value.
Step with alarm clear	Record correlated activity, compare I/Q/magnitude, and seek grounding review. SC/DPS must not recreate device threshold logic.
Stale data, missing samples, gaps, or non-good alarm status	Raise a separate coverage alarm; retain last value/status; recover logging without erasing the file. Apply the activity rules below.
Invalid range, implausible value, clipping, I/Q discontinuity, or frequency mismatch	Mark measurement invalid; check configuration/hardware and escalate to GIZMO expert. It does not establish good isolation.
No alarm during authorized test	Withhold release; verify fixture, threshold, beacon, audible output, SC/DPS, and CT/stimulus path.

Coverage loss: with no ground-affecting activity, record/alarm, preserve data, and recover; unrelated work need not pause solely for stale reporting. With active ground-affecting work, continue only under an approved alternate-monitor procedure, recording source/interval; otherwise pause that activity until coverage or an approved alternative returns. Escalate persistent/recurring loss to both GIZMO and SC/DPS experts.

Investigate recent changes by area/work package. Isolate candidate paths only under the controlling procedure and system owner. Never disconnect protective grounding to clear a report.

9. Shutdown and maintenance

1. Log reason, UTC time, activity state, alarm, quality/range, threshold, configuration, and last healthy sample. Apply coverage rules before removing monitoring.
2. Verify final history write and replica cursors; make any required consistent archive and record checksum/storage reference.
3. Disable excitation with the approved Kria command and independently confirm stimulus-off; stop acquisition with the approved method. Both commands remain TBD in the source.
4. De-energize chassis/PDU only as authorized. Disconnect cables only after power-off and work release. Leave the permanent safety/inductor path in its approved state.

10. ZedBoard spare limits

The spare has its own native OPC UA sidecar, endpoint, and identity with model 1.4.0. It neither proxies nor controls Kria. Installed-loop calibration remains unaccepted: impedance is UncertainLastUsableValue; phase/capacitance are unsupported. Alarm.Active is BadDataUnavailable pending controller/display alarm-bit recovery; its Boolean alone cannot establish clear status.

Anonymous access is read-only. Gated authenticated threshold writes accept 0–1023 Ω ; other writes and operation methods are unsupported. SecurityPolicy None, including password control, restricts use to an isolated maintenance network.

Do not select the spare for monitoring/DPS until calibration, alarm recovery, identity, non-interference, restart, network-loss, security, display comparison, and end-to-end acceptance are complete.

11. Complete before field release

Approve and record: rack/ORC; drawings and safety-assembly topology/IDs; CT orientation/transfer factor/cables; stimulus points; endpoint/trust/roles/network owner; OS/FPGA/runtime/configuration; exact startup/enable/disable/restart/recovery/rollback commands; boot/update/stale/recovery limits; excitation/tolerances; calibration/uncertainty; baseline/threshold/test fixture; SC/DPS fields/alarms; historian/replica/archive retention and restore acceptance; external states/inhibition; pre/post-work and coverage criteria; installed-work procedure; spare acceptance; rack-door revision; Daily Log location; shift and emergency contacts. Unresolved values remain TBD.

Shift record: date/operator; equipment/ORC/configuration/calibration IDs; threshold/excitation; start/end alarm/reason/quality/range/baseline; age/history/replica/archive status; activity/work package/bond windows; contacts; events/interventions; archive/checksum/plots; open issues and handoff.

Source and precedence

[Full guide and source references](#): DUNE DocDB 35392-v2, 1805-v12, 25365-v1, 27482-v2, and 285-v11; legacy engineering/readout reports; maintained Kria, ICD, archive, and Ignition repositories. Released drawings, the current Kria manual, installed characterization, and approved work controls take precedence over prototype or legacy values.